Performance Benchmarking of Embedded Battery State Estimation: A Measurement-Only Comparison of Direct-Measurement, Hybrid, and Data-Driven Estimators

Anonymous

Anonymous,

Abstract

This paper presents a measurement-only performance benchmark for embedded battery state estimation under realistic Battery Management System (BMS) constraints. The study isolates estimator performance rather than hardware-resource or implementation-effort evaluation and compares four embedded-ready estimators on the same Lithium Iron Phosphate (LFP) full-cell trajectory under a shared embedded constraint: a direct-measurement model (DM), a hybrid direct-measurement model with learned State-of-Health support (HDM), a hybrid equivalent-circuit model with voltage feedback (HECM), and a data-driven neural model with GRU-based SOC estimation and LSTM-based online SOH context (DD). The benchmark perturbs only measurable inputs and timing information, including ADC quantization, current bias, additive current, voltage, and temperature noise, missing samples, irregular sampling, burst dropout, voltage spikes, and initial-state mismatch, so that all disturbances remain physically interpretable and reproducible. The comparison reveals no universal winner across deployment-relevant dimensions. Under clean conditions, HDM provides the lowest baseline error with an MAE of 0.0024, followed by HECM with 0.0090, DD with 0.0100, and DM with 0.0311. Under disturbed measurements, HECM is the strongest overall robustness choice, especially in the most discriminative current-bias and voltage-noise scenarios. DD is the most stable estimator under missing samples and recovers fastest after initialization mismatch, while DM shows the expected drift sensitivity and HDM trades nominal accuracy against reduced disturbance tolerance. The benchmark therefore exposes complementary failure modes of direct-measurement, hybrid physics-based, and data-driven estimators and supports decision-oriented model selection for embedded BMS deployment.

Keywords: Battery management system, State of charge, State of health, Performance benchmark, Direct measurement, Equivalent circuit model, Data-driven estimation, Embedded artificial intelligence

---

Nomenclature

Acronyms

BMS	Battery management system
SOC	State of charge
SOH	State of health
DM	Direct-measurement model
HDM	Hybrid direct-measurement model
HECM	Hybrid equivalent-circuit model
DD	Data-driven neural model
ECM	Equivalent circuit model
EKF	Extended Kalman filter
MAE	Mean absolute error
RMSE	Root-mean-square error
ADC	Analog-to-digital converter
CV	Constant-voltage phase
OCV	Open-circuit voltage
GRU	Gated recurrent unit
LSTM	Long short-term memory
LFP	Lithium iron phosphate

Symbols

t	Time
$I(t)$	Current signal at time t
$U(t)$	Voltage signal at time t
$T(t)$	Temperature signal at time t
$\widehat{\mathrm{SOC}}(t)$	Estimated state of charge at time t
$\widehat{\mathrm{SOH}}(t)$	Estimated state of health at time t
$\mathrm{SOC}_{\mathrm{ref}}(t)$	Reference SOC target at time t
$\mathrm{SOH}_{\mathrm{ref}}(t)$	Reference SOH target at time t
ΔI	Current bias or offset
ΔU	Voltage bias or offset
ΔT	Temperature bias or offset
σ_I	Standard deviation of injected current noise
σ_U	Standard deviation of injected voltage noise
σ_T	Standard deviation of injected temperature noise
Q_c	Online cumulative charge state used by the SOC models
C_{nom}	Nominal capacity
C_{eff}	Effective capacity used inside the estimator
$U_{\max}$	Upper voltage threshold used for full-charge detection
U_{tol}	Voltage tolerance used in the full-charge condition
U_1, U_2	Polarization voltages of the first and second RC branch
$\Delta\mathrm{MAE}$	Error increase relative to baseline

1. Introduction

1.1. Motivation and practical relevance

Reliable battery state estimation is a core BMS function because SOC and SOH determine safety margins, usable energy, operating limits, charge control, and degradation-aware system management in both mobile and stationary lithium-ion battery applications [1, 2, 3]. Practical deployment quality is not defined by clean-condition accuracy alone, because real BMS operation is exposed to sensor bias, additive noise, initialization uncertainty after restart, missing samples, irregular sampling, communication interruptions, and transient signal spikes caused by measurement-chain limitations or disturbances in the surrounding system [4, 5, 6]. The relevant engineering question is therefore not only whether an estimator achieves low MAE under nominal measurements, but also whether it converges back to a reliable state after disturbed initialization and whether it remains stable under physically plausible measurement corruption.

As summarized conceptually in Figure 1, the present benchmark treats deployment-facing performance as a three-dimensional problem consisting of nominal accuracy, convergence behavior after state inconsistency, and robustness against disturbed measurements and signal-integrity faults, embedded within broader BMS requirements such as hardware constraints and implementation effort. The present study therefore isolates the performance branch and compares representative estimator classes under identical measurement-side disturbances and a matched embedded-feasibility constraint. The benchmark shows that strong nominal accuracy does not necessarily imply strong disturbance performance and that the estimator classes respond differently to bias, missing samples, and re-initialization events.

1.2. State of the art in battery state estimation

The literature commonly groups battery-state estimators into direct-measurement or integration-based approaches, model-based approaches based on equivalent-circuit or electrochemical models, hybrid approaches that combine physical structure with learned components, and purely data-driven models based on neural sequence learning [7, 1, 2, 3]. Direct-measurement approaches, typically based on Coulomb counting, are attractive because they are simple, computationally inexpensive, and easy to deploy; however, they remain structurally sensitive to current bias, missing data, and initialization errors because they lack an internal correction mechanism [7, 4]. Model-based approaches, especially ECM-

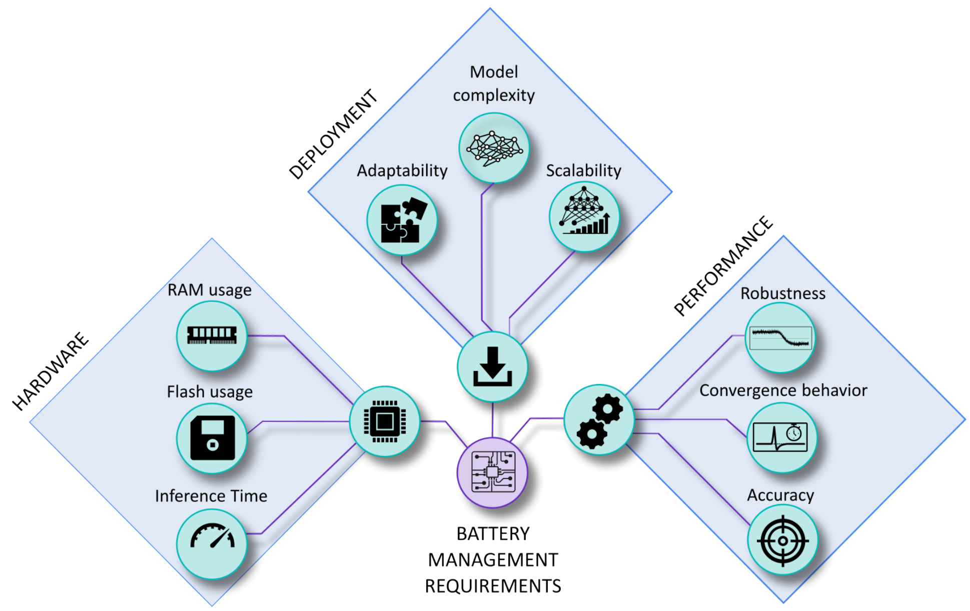

Figure 1: Conceptual overview of deployment-oriented requirements for embedded BMS state estimation. The broader requirement space includes implementation effort, hardware constraints, and estimator performance, whereas the present study focuses on the performance branch and evaluates baseline accuracy, convergence behavior, and robustness under measurement disturbances.

and observer-based estimators, offer physical interpretability and voltage-based correction capability, but their practical behavior depends strongly on the quality of model parametrization, operating-point observability, and measurement quality [8, 2, 4].

Hybrid estimators seek to combine physical state propagation with learned adaptation of capacity, parameters, or auxiliary states, which has made joint SOC/SOH estimation an active research direction [9, 10, 11]. This combination can improve baseline calibration, while it does not necessarily eliminate structural sensitivity to measurement disturbances. Data-driven approaches based on recurrent or sequence models can exploit nonlinear cross-relations between current, voltage, temperature, and derived online features, and recent work has also demonstrated their embedded feasibility through TinyML-oriented implementations, pruning, and quantization [12, 13, 14, 15, 3].

1.3. Performance and embedded deployment gap

A large share of the SOC estimation literature still emphasizes clean-data accuracy, average error metrics, or computational efficiency, while comparatively fewer studies analyze how estimators behave under realistic measurement faults and timing corruption [1, 13, 14, 15]. Existing robustness studies often remain confined to a single model family, for example observer-based robustness against modeling and measurement errors [4] or robust estimation concepts for specific BMS architectures under noise and uncertainty [5], whereas cross-family comparisons under the same disturbance protocol and under comparable embedded constraints remain limited. Benchmarking studies in the broader BMS context underline the importance of reproducible scenario design and deployment-relevant validation, but they do not by themselves close the gap between clean-data estimator comparison and measurement-level performance benchmarking across fundamentally different estimator classes [6]. This gap is especially relevant in embedded BMS design, where estimator selection must jointly consider nominal accuracy, convergence, robustness under disturbed measurements, and computational constraints.

1.4. Contribution and study objectives

This work defines deployment-facing estimator performance as the joint assessment of nominal accuracy, convergence behavior after state inconsistency, and robustness under realistic disturbances applied only to measurable inputs and their timing. The study compares four embedded-ready representatives of major estimator families, namely a direct-measurement model (DM), a hybrid direct-measurement model with learned SOH support (HDM), a hybrid equivalent-circuit model with voltage feedback (HECM), and a data-driven neural model with stateful GRU-based SOC estimation and stateful LSTM-based online SOH context (DD), under identical online execution conditions. The disturbances are deliberately restricted to measurement-level and signal-integrity scenarios, including current bias, current noise, voltage noise, temperature noise, missing samples, irregular sampling, burst dropouts, voltage spikes, ADC quantization, and initialization errors, so that the benchmark remains physically interpretable and experimentally reproducible.

The manuscript should therefore be read as a controlled cross-class estimator benchmark and not as a complete system-level BMS validation campaign. Its role is to isolate estimator-side failure modes under matched data, matched disturbance definitions, matched online execution rules, and matched embedded-feasibility constraints. The resulting contribution is a measurement-only benchmark that separates baseline accuracy, convergence, and disturbed-operation robustness and translates the observed failure modes into decision-oriented guidance for embedded BMS estimator selection.

2. Methodology

2.1. Estimator classes under test

Figure 2 summarizes the causal benchmark chain, from measured input signals and online feature reconstruction to disturbance injection and the final global and local performance evaluation. The direct-measurement model (DM) represents the canonical embedded Coulomb-counting approach with fixed nominal capacity and explicit full-charge reset logic, and serves as the low-complexity reference for integration-driven SOC estimation. All SOH-dependent estimator variants in the benchmark, namely HDM, HECM, and DD, use the same shared LSTM-based online SOH estimator; the classes differ only in how this common SOH signal is consumed inside the SOC pipeline. The hybrid direct-measurement model (HDM) preserves the same Coulomb-counting core but replaces the fixed capacity assumption with an online effective capacity derived from this shared SOH estimator, which makes it representative of lightweight hybrid estimation. The hybrid equivalent-circuit model (HECM) represents physics-based state estimation with observer correction, because it combines current-driven state propagation with voltage-residual feedback and SOH-dependent

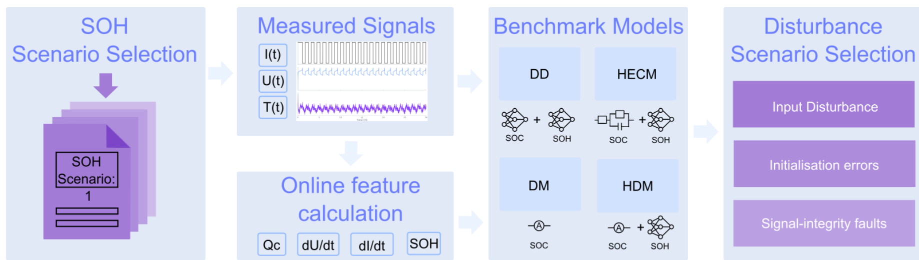

Figure 2: Methodology overview for the causal benchmark chain. Measured battery signals are converted into online auxiliary features, processed by representative estimator classes, exposed to measurement-only disturbances, and finally evaluated through global and local performance metrics.

parameterization in a second-order RC equivalent-circuit structure driven by the same shared SOH estimate [8, 2]. The data-driven model (DD) represents a neural estimator pipeline in which SOC is predicted by a lightweight stateful GRU-based sequence model while the shared LSTM-based online SOH estimate is provided as causal context under the same execution constraints [12, 3].

2.2. SOC estimation methods

2.2.1. Direct-measurement model (DM)

The direct-measurement SOC formulation used in DM, HDM, and as an explicit auxiliary feature in DD follows the same online cumulative-charge state, denoted here uniformly as Q_c. It is updated causally from the measured current stream as

$$Q_c(k) = Q_c(k-1) + \frac{I_k \Delta t_k}{3600\,\mathrm{s\,h^{-1}}}, \tag{1}$$

where I_k is given in ampere and Δt_k in seconds, so that the factor $3600\,\mathrm{s\,h^{-1}}$ converts the increment to ampere-hours. The sign convention is fixed by the benchmark implementation and the state is reset to $Q_c = 0$ after a sustained constant-voltage full-charge event. In the DM, this same charge state is mapped directly to SOC through the fixed nominal capacity

$$\widehat{\mathrm{SOC}}_{\mathrm{DM}}(k) = \mathrm{clip}\left(1 + \frac{Q_c(k)}{C_{\mathrm{nom}}}, 0, 1\right), \tag{2}$$

2.2.2. Hybrid direct-measurement model (HDM)

The HDM preserves exactly the same Coulomb-counting core and differs only in the capacity term. The SOH estimate rescales the usable capacity as

$$C_{\text{eff}}(k) = C_{\text{nom}} \cdot \widehat{\text{SOH}}(k), \tag{3}$$

which yields

$$\widehat{\text{SOC}}_{\text{HDM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{eff}}(k)}, 0, 1\right). \tag{4}$$

This formulation makes the relation between DM and HDM explicit: both use the same online charge state Q_c, while only the denominator changes through online SOH support. In both direct-measurement variants, a sustained near-maximum voltage condition is interpreted as a constant-voltage full-charge event; if the voltage remains above $U_{\text{max}} - U_{\text{tol}}$, where U_{tol} denotes the admissible voltage margin below the upper threshold U_{max}, for the configured CV duration, the internal charge state is reset so that $Q_c = 0$ and therefore $\widehat{\text{SOC}} = 1$, which reflects the practical assumption that a completed CV phase provides a physically anchored full-charge reference.

2.2.3. Hybrid equivalent-circuit model (HECM)

The hybrid ECM model is implemented as a second-order RC model with the state vector

$$\mathbf{x}_k = \begin{bmatrix} \widehat{\text{SOC}}_k & U_{1,k} & U_{2,k} \end{bmatrix}^\top, \tag{5}$$

where U_1 and U_2 denote the polarization voltages of the two RC branches. The prediction step propagates the state according to

$$\mathbf{x}_k^- = \mathbf{A}_k \mathbf{x}_{k-1} + \mathbf{b}_k I_{k-1}, \tag{6}$$

with

$$\mathbf{A}_k = \text{diag}\left(1, e^{-\Delta t_k/\tau_1}, e^{-\Delta t_k/\tau_2}\right), \quad \mathbf{b}_k = \begin{bmatrix} \eta \Delta t_k / C_b \\ R_1\left(1 - e^{-\Delta t_k/\tau_1}\right) \\ R_2\left(1 - e^{-\Delta t_k/\tau_2}\right) \end{bmatrix}, \tag{7}$$

where $C_b = C_0\widehat{\mathrm{SOH}}$ is the SOH-scaled available capacity. The corrected terminal-voltage estimate is written as

$$\widehat{U}_k = \mathrm{OCV}(\widehat{\mathrm{SOC}}_k, \widehat{\mathrm{SOH}}_k, I_k) + U_{1,k} + U_{2,k} + R_i I_k, \tag{8}$$

and the Kalman update uses the residual between $\widehat{U}_k$ and the measured terminal voltage to correct the internal state. In the present implementation, the state propagation is driven by the previous current sample I_{k-1}, whereas the measurement equation uses the current sample I_k available at the correction step. The ECM parameters are not constant. The implementation interpolates R_i, R_1, R_2, τ_1, τ_2, OCV, and dOCV/dSOC from a preidentified lookup table over SOC and SOH, with separate charge and discharge parameter surfaces. The HECM is implemented as a two-RC observer with SOH-dependent parameter scheduling.

2.2.4. Data-driven SOC model (DD)

The data-driven SOC model is a stateful GRU-based recurrent estimator with the feature vector $\{U\ [\mathrm{V}],\ I\ [\mathrm{A}],\ T\ [^\circ\mathrm{C}],\ \mathrm{SOH},\ Q_c,\ dU/dt,\ dI/dt,\ \Delta t\}$. The network consists of a single-layer GRU with hidden size 96 followed by an MLP head with one hidden layer of size 96 and a sigmoid output layer. During optimization, the recurrent core is exposed to causal sequence segments of length

$$L_{\mathrm{SOC}} = 2024 \tag{9}$$

samples. Within this feature set, Q_c is exactly the same online cumulative-charge state defined in (1). It is included explicitly so that the neural estimator receives the same charge-throughput information that also drives DM and HDM. The derivative channels are computed online as finite differences with respect to the current sampling interval,

$$\frac{\mathrm{d}I}{\mathrm{d}t}(k) = \frac{I_k - I_{k-1}}{\Delta t_k}, \qquad \frac{\mathrm{d}U}{\mathrm{d}t}(k) = \frac{U_k - U_{k-1}}{\Delta t_k}, \tag{10}$$

and provide the neural estimator with local transient information on current and voltage dynamics that is not directly captured by the absolute channel values alone. The additional feature Δt_k is passed explicitly so that the network can distinguish amplitude changes from sampling-interval changes under irregular timing. During benchmark execution, the GRU is used as a causal stateful estimator with persistent hidden-state forwarding. No future information is used, and the MLP head is applied at every recurrent step so that a continuous SOC trajectory is obtained directly from the evolving recurrent state, which is consistent with BMS-oriented embedded deployment.

2.3. SOH estimation method

The shared SOH estimator used in the current benchmark is an LSTM-based sequence-to-sequence model that operates on hourly aggregated online features derived from $\{U$ [V], I [A], T [°C], EFC, $Q_c\}$, where the aggregation uses the statistics mean, standard deviation, minimum, and maximum for each feature [16, 17, 18, 19]. This transforms the five base channels into a 20-dimensional hourly feature vector. Architecturally, the SOH network first projects the aggregated input through a two-layer feature embedding block with size 128, layer normalization, GELU activations, and dropout, then processes the sequence with a two-layer unidirectional LSTM of hidden size 160, and finally maps the recurrent output through three residual MLP blocks and a prediction head with hidden size 160 to obtain the SOH estimate for each hourly step.

During optimization, the SOH estimator is exposed to causal hourly sequence segments, and benchmark execution processes one hourly aggregate at a time with causal hidden- and cell-state forwarding. This allows the estimator to accumulate degradation context without access to future information; the first hourly output is anchored with the configured initial SOH value before subsequent hourly predictions are taken directly from the recurrent model output. The SOH prediction is updated at hourly cadence and then held piecewise constant between update points, which enforces an online deployment constraint and avoids sample-wise access to future information while still allowing the faster SOC estimators to use the latest SOH context. In the HDM, the predicted SOH enters the SOC computation through the effective capacity relation $C_{\text{eff}} = C_{\text{nom}} \cdot \widehat{\text{SOH}}$, so the method remains structurally close to direct measurement while replacing the fixed-capacity assumption. In the HECM, the predicted SOH affects the capacity scaling and the lookup-based ECM parametrization, while in the DD model it is provided as an explicit SOC input feature that contextualizes the current measurement trajectory.

2.4. Embedded-oriented model preparation

The focus of the present paper is estimator performance and not hardware-oriented model optimization. Nevertheless, the compared implementations are selected and prepared under the practical requirement that they remain deployable on a common STM32H755ZI-class reference platform with 1 MB flash and 1 MB RAM. This embedded preparation is treated as a prerequisite for fair comparison, so the benchmark does not compare unconstrained large neural models against inherently compact direct-measurement or observer-based estimators. The DM and HECM are already compact because they are based on scalar charge-state updates and lookup-based observer logic. For the neural components, compact deployment-prepared variants are used, namely a stateful SOC-GRU and a stateful SOH-LSTM. These recurrent models correspond to embedded-feasible variants obtained before the present benchmark through structured width reduction, short post-pruning finetuning, and 8-bit weight quantization, yielding hidden size 67 for the SOC-GRU and hidden size 112 for the SOH-LSTM. Table 1 reports the component-level flash and RAM estimates for the reusable building blocks, and Table 2 aggregates them to the estimator level. Together they show that all four benchmarked variants remain within the common reference budget.

Table 1: Estimated persistent flash and RAM footprints of the reusable estimator components for the deployment-prepared variants. For the recurrent neural components, flash estimates refer to the structured-pruned, finetuned, and 8-bit quantized models. RAM* includes only architecture-intrinsic persistent numerical states; temporary buffers, framework overhead, stack usage, and system-level runtime memory are excluded.

Component	Flash [KiB]	RAM* [B]	Basis
DM core (Coulomb counting)	4.2	16	four persistent scalar states in the online update logic
SOC-GRU core	27.9	268	one recurrent hidden state with size 67 stored as float32
SOH-LSTM core	408.3	1792	two-layer LSTM with hidden and cell states of size 112 stored as float32
HECM core (2RC observer)	442.5	52	observer state $x \in \mathbb{R}^3$, covariance $P \in \mathbb{R}^{3\times3}$, and previous current

Table 2: Estimated combined estimator footprints relative to a reference embedded budget of 1 MB flash and 1 MB RAM. The recurrent neural components correspond to the deployment-prepared variants after structured pruning, finetuning, and 8-bit quantization. RAM* reports only persistent numerical states and therefore serves as a strict lower bound rather than as a full runtime-memory measurement.

Estimator	Flash [KiB]	RAM* [B]	Flash $<$ 1 MB	RAM* $<$ 1 MB
DM	4.2	16	yes	yes
HDM	412.5	1808	yes	yes
HECM	850.8	1844	yes	yes
DD	436.2	2060	yes	yes

2.5. Performance benchmark design

The benchmark follows a strict measurement-only principle: only quantities that a real BMS measures or derives online from measured data are manipulated, namely current, voltage, temperature, sample availability, and sampling time, whereas artificial parameter mismatch and synthetic hidden-state corruption are deliberately excluded [6]. Figure 3 groups the benchmark into input disturbances, initialization errors, and signal-integrity faults, while Table 3 defines the exact disturbance implementations used in the study.

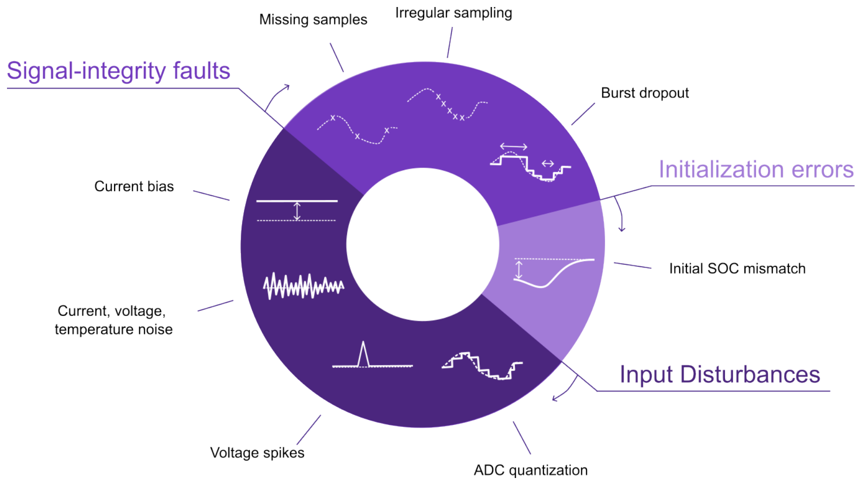

Figure 3: Taxonomy of the disturbance scenarios used in the performance benchmark, grouped into input disturbances, initialization errors, and signal-integrity faults.

The disturbance magnitudes are not intended as one-to-one copies of a single field specification. Instead, they were selected from three justification classes: deployment-level sensing limitations, adverse measurement corruption, and deliberate signal-integrity stress tests [20, 21, 22, 23, 24, 25]. The main benchmark matrix follows the scenario families shown in Figure 3 and provides the basis of the main result figures and tables. ADC quantization is part of the same scenario set and is included in the cross-scenario overview; its detailed numerical results are reported in Appendix Table B.9 and Figure B.14. Overall, the benchmark is designed as a controlled and reproducible scenario study that exposes estimator-specific failure modes under a common disturbance protocol rather than emulating the complete system stack of a production BMS.

Table 3: Disturbance configurations used in the benchmark and their practical rationale. Exact numeric values are fixed benchmark settings; the cited sources motivate disturbance type and magnitude range.

Scenario	Benchmark setup	Practical origin / rationale
ADC quantization	$\Delta I = 0.01\,\mathrm{A}$; $\Delta U = 0.005\,\mathrm{V}$; $\Delta T = 0.5\,^{\circ}\mathrm{C}$	ADC steps; sensor scaling; rounding; coarse deployment-level discretization [20, 21, 26]

Table 3: Disturbance configurations used in the benchmark and practical rationale (continued).

Scenario	Benchmark setup	Practical origin / rationale
Current bias	Constant $+0.05\,\mathrm{A}$; sweep: 0.5%, 1.5%, 3.0% of $\lvert I\rvert_{\max}$	Sensor offset; gain error; miscalibration; drift; upper bound as stress case [21, 22, 23]
Current noise	Gaussian; $\sigma_I = 0.10\,\mathrm{A}$; local detail up to $\sigma_I = 0.20\,\mathrm{A}$	Sensing-chain noise; EMI pickup; stronger local stress sweep [4, 5]
Voltage noise	Gaussian; $\sigma_U = 0.01\,\mathrm{V}$	Wiring noise; front-end noise; poor filtering; EMI [4, 20, 24]
Temperature noise	Gaussian; $\sigma_T = 1.0\,^{\circ}\mathrm{C}$	Sensor tolerance; conversion noise; thermal gradients; sensor-to-cell mismatch [26, 11]
Initial SOC error	Initial mismatch: -0.10 SOC; DD via perturbed online Q_c	Restart; storage; missing recent history; weak LFP OCV anchoring [4, 8]
Missing samples	Every 50th sample frozen; every 50 s	Packet loss; blocked updates; logger dropouts; periodic data freeze [5, 25]
Irregular sampling	Uniform jitter: $\pm 0.1\,\mathrm{s}$ around nominal 1 s grid	Scheduling jitter; bus latency; asynchronous acquisition [6, 5, 25]
Burst dropout	One central frozen gap: 3600 s	Communication interruption; frozen task; stuck data stream; severe recovery case [6, 5]
Voltage spikes	Single-sample spikes: $\pm 0.20\,\mathrm{V}$ every 1000 samples (1000 s)	EMI; switching disturbance; corrupted voltage samples [24]

2.6. Metrics and evaluation criteria

For deployment-oriented interpretation, the benchmark results are synthesized along three performance dimensions: nominal accuracy under baseline conditions, convergence behavior after initialization mismatch or interrupted signal continuity, and robustness under sustained or transient measurement corruption. Disturbance-related global performance is quantified through MAE, RMSE, bias, maximum error, and the 95th percentile absolute error, because these metrics provide a compact view of average performance, systematic deviation, and tail behavior across the full trajectory. Local disturbance performance is analyzed through scenario-specific metrics such as jump counts, output variance, absolute-error variance, and excess rolling MAE, because several disturbances do not primarily manifest as full-run MAE collapse but as short-term output fluctuations, delayed drift, or slow re-synchronization. Whenever a disturbance mask exists, the analysis further separates pre-disturbance, disturbed, and post-disturbance behavior in order to quantify disturbance-window error growth, recovery time, and residual post-disturbance error.

In the present benchmark phase, ADC quantization, current bias, voltage noise, temperature noise, missing samples, and irregular sampling are interpreted primarily through global metrics, whereas burst dropout, voltage spikes, current-noise detail, and initialization mismatch are additionally interpreted through dedicated local evidence because their most relevant effects are episodic or recovery-related. Recovery after initialization errors is interpreted through two baseline-relative operating bands rather than through one fixed universal threshold. This keeps the recovery criterion tied to the normal clean-condition performance level of each estimator instead of imposing the same absolute tolerance on models with very different baseline accuracy. The strict band is defined as $\max(\mathrm{P95}_{\mathrm{base}}, 1.5 \cdot \mathrm{MAE}_{\mathrm{base}})$ so that recovery means re-entry close to the normal baseline regime while avoiding unrealistically narrow thresholds. The use of $\mathrm{P95}_{\mathrm{base}}$ protects against heavy-tail baseline errors, whereas the $1.5 \cdot \mathrm{MAE}_{\mathrm{base}}$ term prevents the threshold from being governed by isolated tail behavior alone. The fair band is defined as $\max(2 \cdot \mathrm{P95}_{\mathrm{base}}, 3 \cdot \mathrm{MAE}_{\mathrm{base}}, 0.02)$ and serves as a more deployment-tolerant recovery criterion for estimators that re-synchronize to a practically acceptable level without necessarily returning immediately to their strict near-baseline range. The lower bound of 0.02 avoids unrealistically tight fair-band thresholds for very accurate baseline models. Both recovery bands require sustained re-entry over a fixed time horizon so that the reported recovery time reflects stable re-synchronization rather than a short accidental crossing of the threshold. Using both a strict and a fair band also reduces dependence on any single heuristic threshold choice. This multi-scale metric design evaluates performance as a structured combination of nominal accuracy, convergence behavior, global degradation, local disturbance response, and long-term drift.

3. Experimental Setup

3.1. Dataset and data acquisition

The benchmark is based on the MGFarm LFP 18650 dataset, which was generated in a laboratory environment as a controlled design-of-experiments study on cyclic aging and operating conditions and provides a suitable basis for performance analysis because it combines long full-cell trajectories with high-resolution measured signals and repeated checkup structures [27, 28, 29]. The current manuscript phase uses one representative full-cell trajectory from this dataset in order to isolate estimator behavior under controlled measurement corruption before extending the analysis to additional cells. The available raw measurements comprise current, voltage, temperature, and test time, while the feature-engineered dataset also contains offline reference targets and derived variables such as capacity, SOH, equivalent full cycles, and cumulative charge state. This setup is well suited for the present study because it combines realistic laboratory measurements with enough structure to derive reproducible disturbance scenarios without introducing external domain shift.

3.2. Reference targets and preprocessing

The reference preprocessing first computes the signed transferred charge as $Q_k = I_k \Delta t_k / 3600$ and the absolute transferred charge as $|Q_k|$, from which the capacity during dedicated capacity-test intervals is obtained as $C_{\text{ref}} = \sum_k |Q_k|$. The reference SOH target is then defined offline as $\text{SOH}_{\text{ref}} = C_{\text{ref}} / C_{\text{ref},0}$, where $C_{\text{ref},0}$ denotes the first measured reference capacity of the trajectory, and the resulting capacity and SOH columns are interpolated between the dedicated capacity-test anchors. The reference SOC target used in the benchmark corresponds to the offline Coulomb-counting target $\text{SOC}_{\text{ref}} = 1 + Q_c / C_{\text{ref},0}$, where the cumulative charge state Q_c is propagated from the signed charge and reset to 0 at the upper voltage anchor and to $-C_{\text{ref},0}$ at the lower voltage anchor during preprocessing. These reference targets are intentionally treated as offline evaluation quantities, because they rely on dataset-side processing and anchor logic that are not fully available to an online estimator during deployment.

3.3. Test trajectory and benchmark scenarios

All estimators are evaluated on the same benchmark trajectory and with the same disturbance protocol so that performance differences can be attributed to estimator structure under matched data and target definitions. The reported benchmark scenarios comprise the nominal baseline, current bias, additive current noise, voltage noise, temperature noise, initial SOC error, missing samples, irregular sampling, burst dropout, and voltage spikes, which together cover both measurement corruption and signal-integrity faults; the exact disturbance magnitudes and timing rules are summarized in Table 3. Input disturbances are applied directly to the measured channels, initialization errors are injected only where structurally meaningful, and signal-integrity scenarios manipulate sample availability or timing while preserving the online execution logic of the estimators. The scenario set is designed to expose different failure mechanisms, including long-term integration drift, loss of voltage-based correction, transient overshoot, stability under frozen inputs, and the ability to recover after local disturbances.

3.4. Implementation details

Every estimator is executed under online conditions, meaning that the disturbed measurement stream is processed sample by sample and all auxiliary quantities required by the models are rebuilt directly from the disturbed inputs. This applies in particular to the derived online features Q_c, EFC, dU/dt, dI/dt, and the hourly SOH aggregates, so that no hidden dataset-side variables leak into the benchmark runs. For the recurrent neural estimators, the intended embedded execution mode is explicitly stateful. The SOH-LSTM forwards hidden and cell states across successive hourly updates, and the SOC-GRU forwards its hidden state causally along the measurement stream. In addition, the neural variants used for deployment preparation follow the compact pruned-and-quantized architecture described above, so that the benchmarked estimator classes remain compatible with the reference embedded memory budget. In freeze and dropout scenarios, the disturbance propagation is handled consistently by freezing or distorting the derived online features together with the corresponding measured inputs, which is necessary for a physically meaningful comparison between model classes. The direct-measurement estimators include the same CV-triggered reset mechanism that is also used to define a physically anchored full-charge point in preprocessing, whereas the data-driven initial-SOC test is implemented through an initial offset of the online Q_c feature because the neural model does not expose an explicit SOC state variable. The simulation environment enforces identical nominal capacity assumptions, disturbed input streams, evaluation targets, and global/local metrics across all estimators, which makes the resulting comparison directly attributable to estimator design under matched execution conditions.

4. Results

4.1. Baseline performance

Figure 4 summarizes the nominal baseline performance on the selected benchmark trajectory. The hybrid direct-measurement model (HDM) achieves the lowest baseline error with MAE = 0.0024, followed by the hybrid equivalent-circuit model with 0.0090, the data-driven model with 0.0100, and the direct-measurement model with 0.0311, making HDM the clean-condition leader. This baseline ranking serves as the reference point for the following disturbed-measurement scenarios and for the corresponding robustness comparisons; the complete baseline metrics are listed in Appendix Table A.5.

4.2. Current-bias sensitivity

Figure 5 summarizes the current-bias results and includes in panel (a) a representative example of the applied +3% bias; the exact bias protocol is listed in Table 3. Across the matched bias sweep (0.5%, 1.5%, and 3.0% of $|I|_{\max}$), the hybrid equivalent-circuit model (HECM) remains the most robust estimator, the data-driven model (DD) is intermediate, and both direct-measurement variants

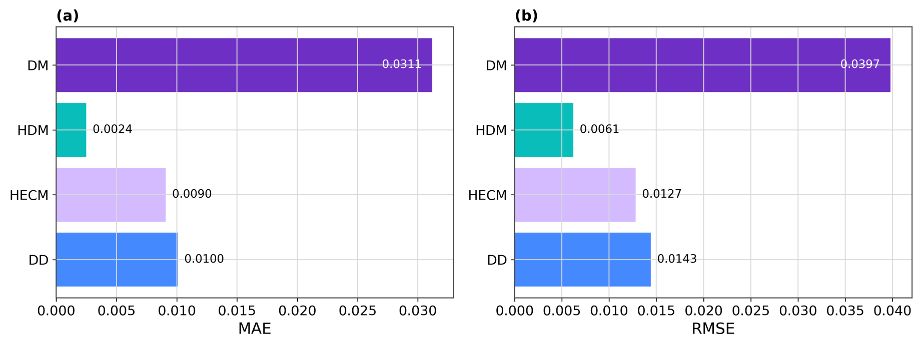

Figure 4: Baseline performance on the selected full-cell benchmark trajectory. The hybrid direct-measurement model provides the lowest clean-condition error, whereas the direct-measurement model shows the largest baseline deviation.

degrade strongly. In the reported current-bias scenario, HECM reaches MAE = 0.0932, compared with 0.2859 for DD, 0.3068 for HDM, and 0.3143 for DM. The detailed numerical results for the bias scenario are included in Appendix Table A.6.

4.3. Noise sensitivity

Current noise is weaker than current bias in global MAE, but it remains informative through local behavior and output variability. In the compact matrix, the reported high-noise case uses $\sigma_I = 0.10\,\mathrm{A}$, while the local detail sweep in Figure 6 extends the current-noise axis up to $\sigma_I = 0.20\,\mathrm{A}$. Under the high-noise case, the hybrid equivalent-circuit model shows the largest global penalty ($\Delta\mathrm{MAE} = 0.0075$), whereas DM, HDM, and DD remain close to their baseline MAE values. The local noise story differs from the global metric: the data-driven model exhibits the strongest output variability because current noise propagates into several derived input channels, whereas the hybrid equivalent-circuit model is the most sensitive in the error domain. The broader noise block is not limited to current noise: temperature noise with $\sigma_T = 1.0\,^\circ\mathrm{C}$ is almost neutral for DM and HECM but increases the error of HDM and DD, while voltage noise with $\sigma_U = 0.01\,\mathrm{V}$ is one of the strongest discriminators because HECM remains nearly unaffected whereas DM, HDM, and DD de-

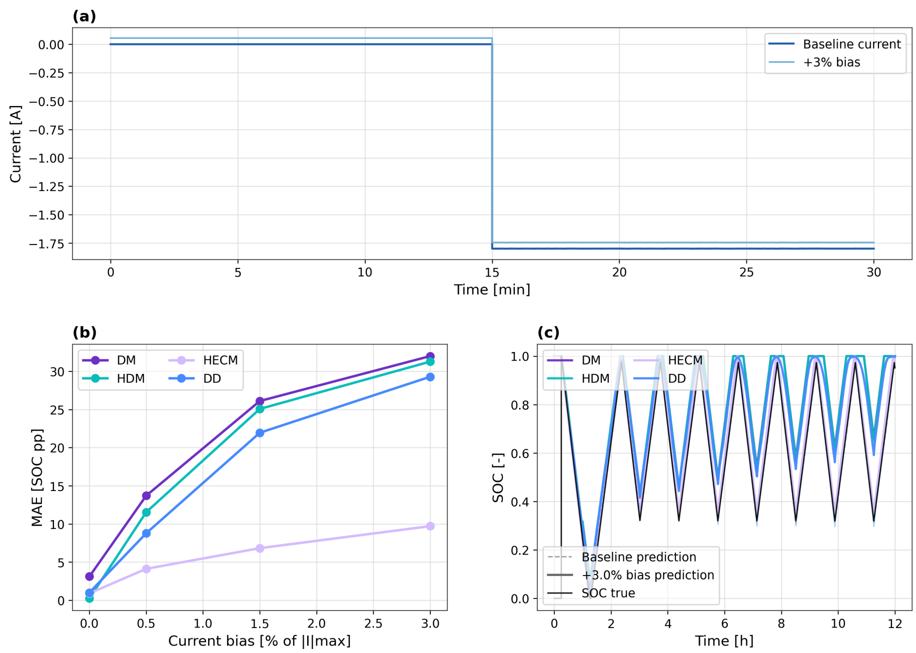

Figure 5: Sensitivity to current measurement bias. (a) Example of the applied current bias (+3%) in a representative 30 min window. (b) Global degradation of SOC estimation accuracy as a function of bias magnitude. (c) SOC trajectory divergence over the first 12 h for a +3% current bias.

grade strongly. Figure 6 therefore separates the applied current disturbance, a matched SOC comparison in the same window, the global MAE penalty, and the local output-variability summary, while Appendix Tables A.6 and A.7 provide the complementary numerical evidence for current, voltage, and temperature noise.

4.4. Initialization error and recovery

Initialization performance must be analyzed locally and in an evaluation window without constant-voltage reset events, because global MAE hides the actual re-synchronization behavior after a wrong initial SOC. The injected mismatch in the benchmark is -0.10 SOC, as summarized in Table 3. DM recovers to the strict baseline band after about 1.17 h, HECM recovers after about

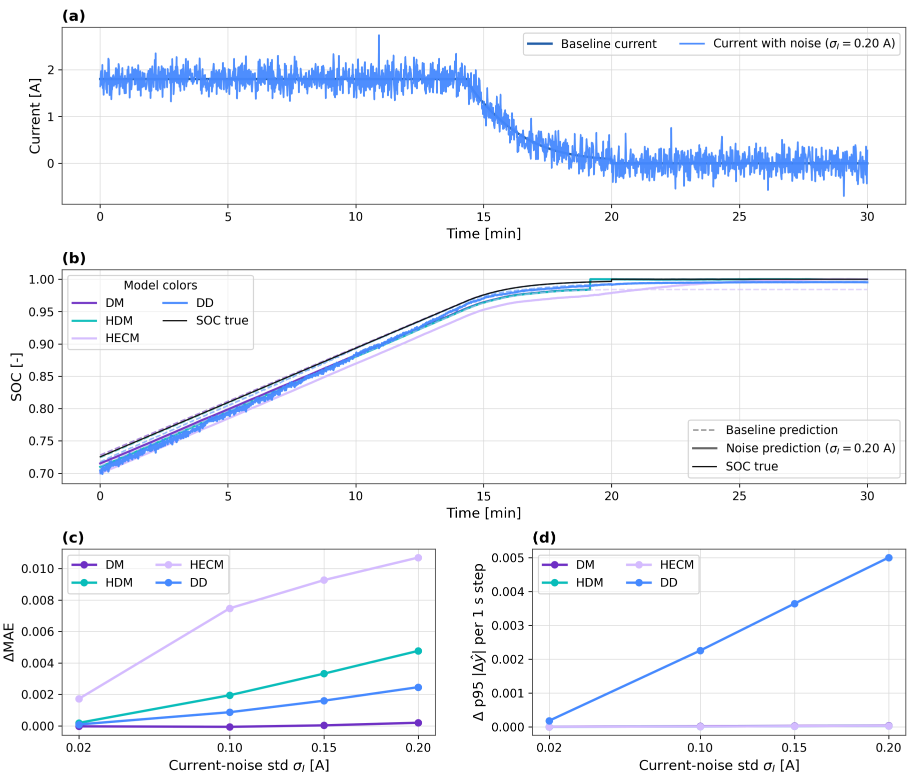

Figure 6: Noise sensitivity under additive current noise. The figure combines an applied current-noise example, a matched SOC comparison in the same 30 min window, the global MAE penalty, and the local output-variability sensitivity across noise levels.

2.37 h under the strict criterion and after about 1.16 h under the fair criterion, whereas HDM does not re-enter either band in the inspected window. The data-driven model is evaluated through an initial perturbation of the online Q_c feature rather than through an explicit hard SOC-state reset, which keeps the test aligned with deployment-time observables; under this formulation it recovers after about 1.05 h under the strict criterion and after about 1.00 h under the fair criterion. Figure 7 visualizes the resulting reset-free recovery behavior, and the corresponding recovery metrics and baseline-band thresholds are listed in Appendix Table A.7.

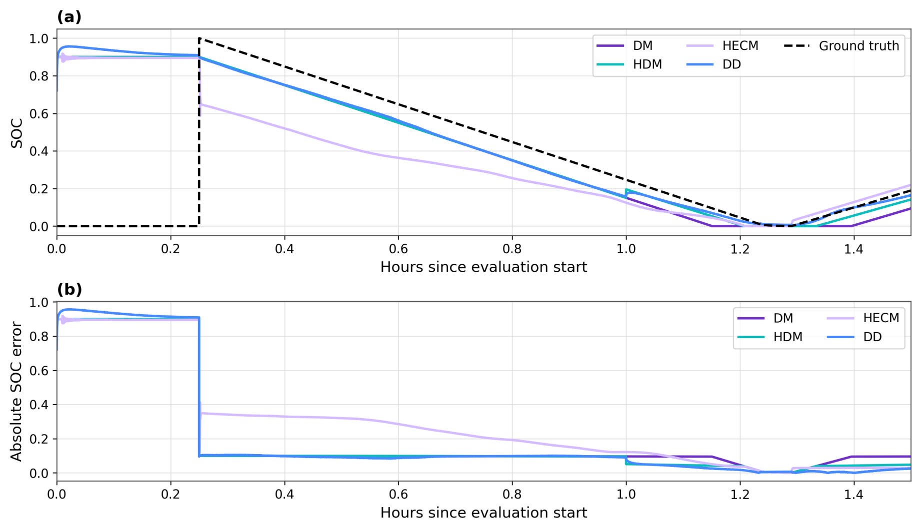

Figure 7: Initial-state recovery in a reset-free evaluation window. The top panel shows SOC trajectories after the imposed state mismatch, and the lower panel shows the corresponding absolute SOC error over time.

4.5. Missing samples and signal integrity

Missing samples are the strongest global signal-integrity stressor in the benchmark. In this scenario, every 50th sample is frozen, corresponding to one held sample every 50 s on the nominal 1 s grid; under this protocol the data-driven model changes only marginally from 0.0100 to 0.0117 MAE, whereas DM and HDM degrade much more strongly. The irregular-sampling scenario jitters the sampling intervals uniformly by ± 0.1 s around the nominal 1 s grid while preserving the full-run duration. Under this resampled implementation it remains nearly neutral for all models, with only very small signed MAE changes around zero. The burst-dropout scenario consists of one central frozen gap of 3600 s. It remains globally moderate, but all models require roughly 27–29 h to return to the baseline error band, making the post-gap recovery behavior more informative than the global MAE alone. The signal-integrity block therefore needs both global and local views: Figure 8 summarizes the global penalties for missing samples, irregular sampling, and burst dropout, Figure 9 shows the transition

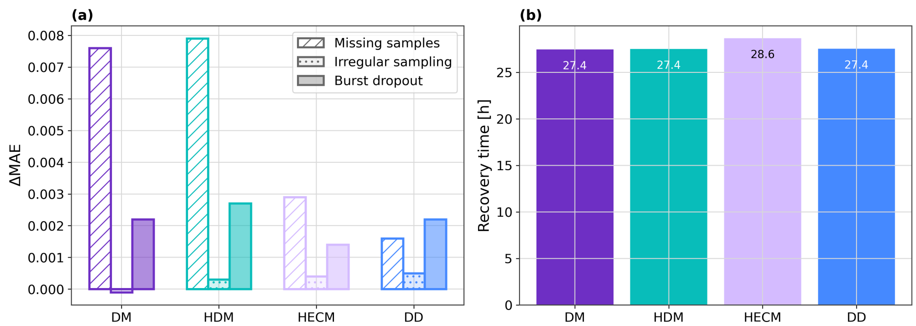

Figure 8: Signal-integrity performance summary. The left panel compares the global MAE penalties for missing samples, irregular sampling, and burst dropout, whereas the right panel summarizes the burst-dropout recovery time.

into the gap, and Figure 10 shows the long recovery afterwards. The corresponding global metrics are listed in Appendix Table A.6, the burst-dropout recovery times in Appendix Table A.7, and the scenario-to-evidence mapping in Appendix Table A.8.

4.6. Spike sensitivity

Single voltage spikes have very different local effects across model classes, even when the full-run MAE changes remain small for DM, HDM, and HECM. The spike protocol injects one $\pm 0.20\,\mathrm{V}$ voltage spike every 1000 samples, corresponding to every $1000\,\mathrm{s}$ on the nominal $1\,\mathrm{s}$ time base. The data-driven model shows the clearest spike sensitivity, both in the representative SOC window and in the local aligned error response, with a noticeable global increase from 0.0100 to 0.0144 MAE in the present spike protocol. Figure 11 summarizes this behavior through a representative local window, a spike-severity summary, and the aligned post-spike error response. To avoid misleading signed full-run averages, the compact spike summary is reported as the 95th percentile of the post-spike peak excess error rather than as signed ΔMAE; the numerical spike-related metrics are summarized in Appendix Tables A.6 and A.7, and the role of local evidence for spikes is highlighted in Appendix Table A.8.

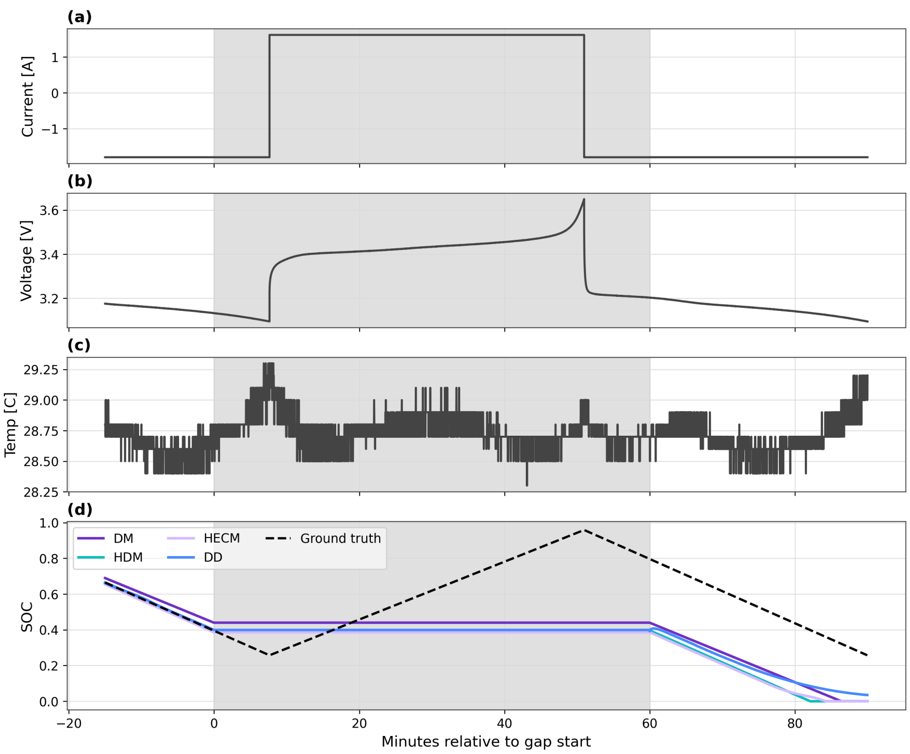

Figure 9: Local transition into the burst-dropout window. The shaded region marks the missing-data interval, and the lower panel shows how the estimators behave while the measurements are frozen.

4.7. Cross-scenario comparison

Figure 12 condenses the disturbed-scenario penalties into one cross-scenario view and makes the estimator-specific pattern shifts directly visible; it also includes the ADC-quantization case. The cross-scenario view confirms that the benchmark does not yield one universal winner across all criteria, because HDM is best under nominal conditions, HECM is most resilient under current bias and voltage noise, DD is the most stable estimator under missing samples, and DD also shows the fastest strict-band recovery after initialization mismatch. Current bias and voltage noise are the most discriminative scenarios in the present benchmark phase because they most clearly separate correction-capable estimators from purely integration-driven or fully data-driven alternatives. The

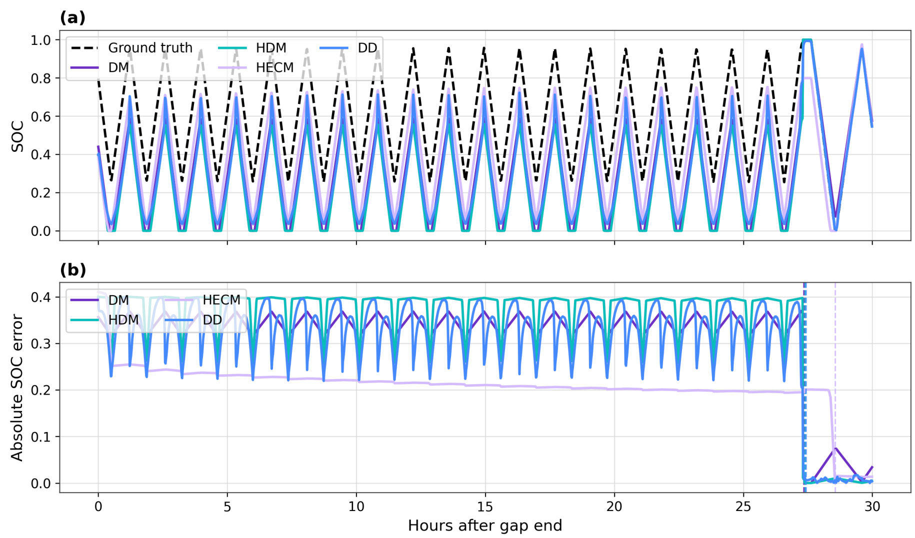

Figure 10: Post-gap recovery after the burst dropout. The upper panel shows the SOC trajectories after the gap, and the lower panel shows the corresponding absolute SOC error together with the recovery markers.

heatmap is useful because it shows where the ranking changes across disturbance classes and where estimator strengths are scenario-specific rather than universal; the corresponding numerical overview is compiled in Appendix Tables A.5–A.7.

5. Discussion

5.1. Model-specific performance behavior

The direct-measurement model acts as the canonical drift-prone estimator, because every persistent current disturbance directly accumulates in the SOC estimate and missing measurement information cannot be corrected by an additional feedback mechanism. The hybrid direct-measurement model improves nominal calibration through online capacity adaptation, but it remains structurally exposed to the same current-integration failure mode and additionally inherits the dynamics of the learned SOH support. The hybrid equivalent-circuit model, combined here with the same shared data-driven online SOH estimator

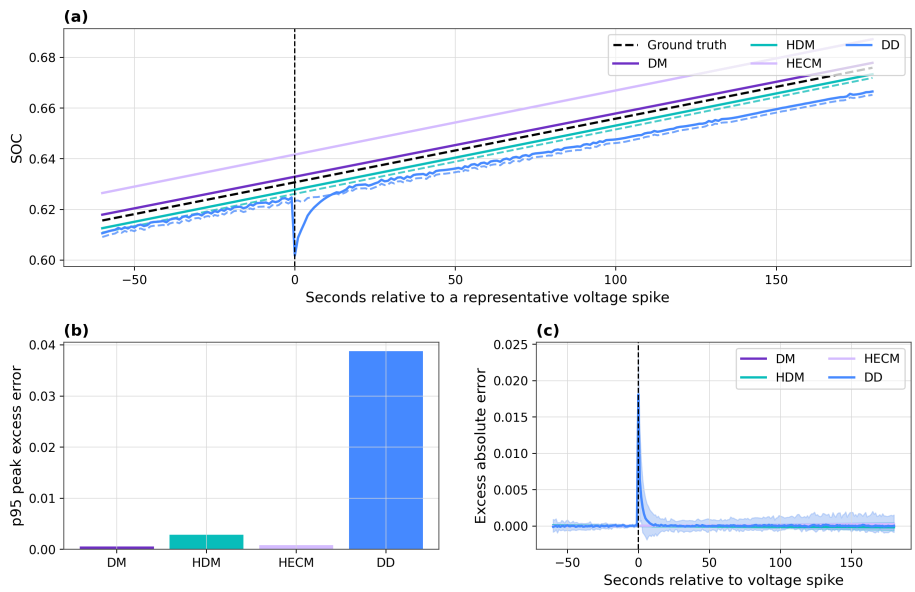

Figure 11: Voltage-spike sensitivity. The figure combines a representative SOC window around a spike, a local spike-severity summary based on the 95th percentile of the post-spike peak excess error, and the aligned local excess-error response across repeated spikes.

used by the other SOH-aware variants, benefits from interpretable ECM parameterization, explicit voltage-residual correction, and causal observer structure. This combination explains why it handles current bias and voltage noise much better than the integration-based alternatives, even though it becomes more sensitive when noise perturbs the current-dependent parameter selection and observer update.

The present data-driven neural model behaves as a compact context-rich estimator whose robustness depends strongly on the disturbance channel: it is notably stable under missing samples, competitive under current bias, and the fastest model after initialization mismatch, but more exposed to derivative-amplified noise and voltage spikes. Read together, these results suggest that the current compact instance still relies too strongly on dominant trigger features from the input stream and does not yet exploit voltage and temperature information as effectively as the best hybrid physics-based alternative. That

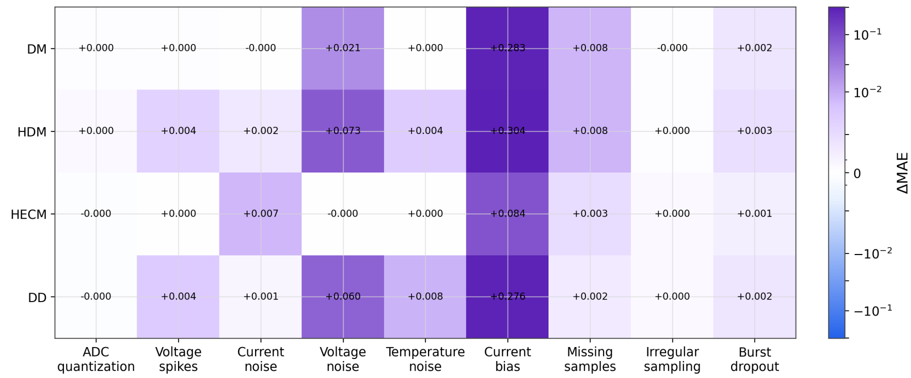

Figure 12: Cross-scenario error increase relative to the baseline case. The heatmap summarizes the disturbed-scenario ΔMAE values, including ADC quantization, and highlights where robustness rankings change across disturbance classes.

observation should be read as a design lesson rather than as a class limitation. Unlike the fixed-form direct-measurement and ECM estimators, the data-driven class retains a comparatively large tuning space through architecture, retraining, feature selection, and deployment-time state handling, so its present benchmark ranking reflects one compact embedded instance rather than a hard ceiling of the class itself.

5.2. Accuracy–performance trade-offs

The benchmark makes clear that low baseline MAE does not imply robust disturbed operation, because the best nominal model under clean measurements is not the best model under current bias or voltage noise. The observed trade-off is therefore not only between accuracy and computational complexity, but also between calibration strength, explicit physical correction capability, recovery speed after state inconsistency, and sensitivity to disturbance propagation through learned or derived features. The benchmark consequently yields a structured performance split: the hybrid direct-measurement model is the nominal-accuracy leader, the hybrid equivalent-circuit model is the overall robustness leader, and the data-driven neural model converges back to the strict baseline band fastest after initialization mismatch. This result argues against single-metric leaderboard thinking and favors a scenario-aware interpretation in which estimator quality is tied to the disturbance classes and recovery requirements that matter most in the target application.

5.3. Implications for BMS deployment

If current-sensor bias and long-term drift are dominant concerns, the benchmark evidence favors the hybrid equivalent-circuit approach, because the observer can re-anchor the SOC estimate through the measured voltage while the shared online SOH model keeps the parameterization aging-aware. If the deployment focus is nominal accuracy under controlled sensing and low-compute execution, the hybrid direct-measurement class remains attractive, but its vulnerability to biased current measurements must be considered explicitly. If rapid recovery after restart or initial-state mismatch is the primary requirement, the data-driven neural approach is attractive because it re-enters the strict baseline band fastest in the present benchmark, while simultaneously remaining strong under missing-sample stress. If robustness against signal-integrity issues such as missing samples is important, the data-driven model offers a strong alternative, provided that the input feature pipeline is designed so that the estimator does not over-focus on a single dominant measurement channel and remains stable under voltage-driven disturbances.

While the present study focuses on performance, practical BMS deployment also requires consideration of implementation complexity and scalability when the observed estimator behavior is transferred to real systems. Within that broader design space, the data-driven class retains the clearest improvement headroom because architecture, features, and training can be updated directly, whereas ECM-based estimators are more constrained by model topology and parameter-identification workflow. At the same time, the hybrid equivalent-circuit configuration used here shows why a physics-based SOC core plus shared data-driven SOH support can already yield a very strong compromise between interpretability, disturbance correction, and embedded feasibility. The footprint estimates indicate that persistent state memory is negligible for all estimators in the present study and that all four deployment-prepared estimator variants remain below the reference 1 MB flash budget. The tightest flash case is the hybrid equivalent-circuit estimator because it combines the ECM lookup structure with the shared SOH-LSTM, whereas the direct-measurement baseline keeps the largest margin. For embedded BMS design, the practical selection criterion should therefore combine the expected fault profile of the measurement chain with implementation constraints. Table 4 condenses the benchmark outcomes into a decision-oriented recommendation map so that the reader is not left with a large set of disconnected scenario results.

Figure 13 complements this recommendation map with an illustrative visual synthesis of the same benchmark evidence. The values in this figure are relative scores rather than additional physical measurements. For each lower-is-better metric, the four estimator classes are min–max normalized as $(x_{\max} - x)/(x_{\max} - x_{\min})$, so that the best class in the present comparison receives 1 and the weakest class receives 0. The accuracy dimension averages the normalized baseline MAE, RMSE, and 95th-percentile error. The robustness dimension averages the normalized ΔMAE values across the disturbed scenarios listed in Appendix Table A.8. The recovery dimension uses the strict recovery time after the initial-SOC perturbation; missing recovery within the inspected window is treated as non-recovery and assigned the lowest score. The bar chart then applies three illustrative priority weightings to these three dimensions: accuracy-weighted (0.60/0.20/0.20), robustness-weighted (0.20/0.60/0.20), and recovery-weighted (0.20/0.20/0.60), where the entries correspond to accuracy, robustness, and recovery, respectively. Thus, each profile assigns 60% of the composite score to its focal dimension and 20% to each of the other two dimensions. The accuracy-weighted profile should therefore be read as an accuracy-prioritized compromise, not as the pure baseline-accuracy ranking reported in Table 4. The synthesis is intended as a compact orientation aid and does not replace the raw scenario-wise results.

Table 4: Decision-oriented recommendation map derived from the benchmark results. The table summarizes which estimator class is most suitable when one deployment priority dominates the design decision.

Primary deployment priority	Recommended class	Main supporting benchmark evidence	Main trade-off to consider
Lowest nominal SOC error under clean sensing	HDM	Lowest baseline MAE on the benchmark trajectory (0.0024)	Strong sensitivity to current bias and no recovery to the baseline bands in the inspected initialization-mismatch window
Best overall robustness under uncertain measurement quality	HECM	Strongest current-bias robustness, nearly unchanged under voltage noise, and competitive baseline accuracy	Larger flash footprint and slower strict-band recovery after initialization mismatch than DD
Fastest re-synchronization after restart or wrong initial state	DD	Fastest strict-band recovery in the local initialization analysis	More sensitive than HECM to voltage spikes and less robust than HECM under current bias
Highest resilience to missing samples and signal freezing	DD	Smallest MAE increase under missing samples and strong post-disturbance stability	Clean-condition accuracy remains below HDM, and voltage-driven disturbances require careful feature-pipeline design
Best development headroom for iterative model improvement	DD	Architecture, features, and training can be updated directly while retaining embedded-feasible deployment and strong performance in recovery and missing-sample scenarios	Present compact instance is still weaker than HECM in the strongest bias- and voltage-driven stress cases
Simplest implementation and largest resource margin	DM	Minimal architecture and smallest flash footprint by a wide margin	Weakest robustness under biased current measurements and missing-sample stress

5.4. Limitations and future extensions

The current benchmark phase is based on one representative dataset trajectory and should later be expanded to additional cells, broader aging states, and cross-cell validation before stronger generalization claims are made. The spike study currently uses isolated spike events; clustered disturbances, longer spike sequences, and hardware-level disturbance injection may reveal additional model differences. The initialization test for the data-driven model remains an online-state proxy based on Q_c perturbation rather than an explicit latent-state reset, which is methodologically justified but should still be interpreted as a class-specific approximation. Future extensions should include the remaining extended scenarios, additional cells, pack-level settings, and eventually

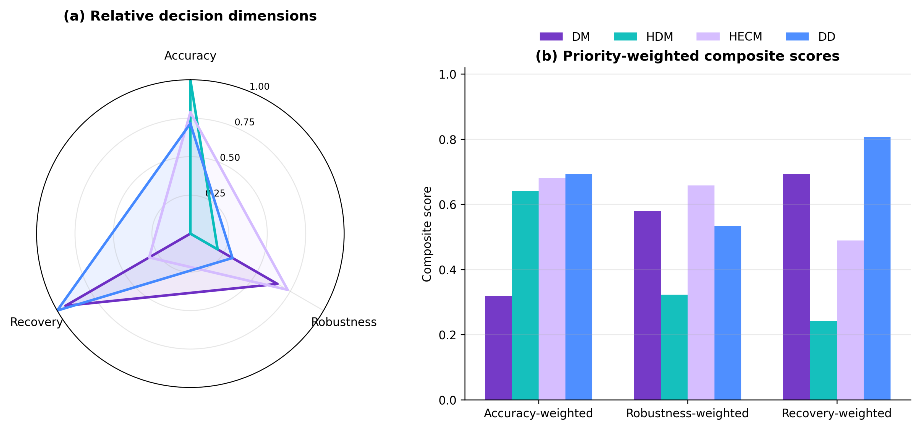

Figure 13: Decision-oriented synthesis of the benchmark results. Panel (a) collapses the benchmark into three relative, min–max-normalized decision dimensions, namely nominal accuracy, disturbed-scenario robustness, and recovery behavior. Panel (b) shows three illustrative deployment-priority profiles and is intended only as a visual complement to Table 4, not as a replacement for the raw scenario metrics.

hardware-in-the-loop or true embedded runtime benchmarks under the same performance-oriented evaluation philosophy.

6. Conclusion

This work frames battery-state estimator evaluation as a performance problem under realistic measurement corruption rather than as a clean-data accuracy comparison alone. The central engineering message is that estimator selection depends on the expected disturbance profile of the BMS measurement chain, the required recovery behavior after corrupted inputs, and the practical implementation constraints of the target system, because no single estimator dominates all deployment-relevant dimensions. The main findings can be summarized by estimator family: the hybrid direct-measurement model is the nominal-accuracy leader, the hybrid equivalent-circuit model with shared data-driven SOH support is the strongest overall robustness choice in the most discriminative disturbance scenarios, and the data-driven neural model is strongest in recovery after initialization mismatch and in missing-sample resilience.

If one general-purpose estimator must be preferred under the specific disturbance profile emphasized here, the benchmark supports the hybrid equivalent-circuit approach as the strongest overall compromise. Its combination of interpretable physical parameterization, voltage-based correction, and compact shared SOH support preserves a clear role for ECM-based estimation in embedded BMS design. At the same time, the data-driven model class remains highly attractive when the development strategy includes iterative data collection and model updates. The present benchmark also yields a practical design lesson: data-driven SOC estimators should not rely too strongly on one dominant trigger variable, but should be trained and feature-engineered to make balanced use of the available current, voltage, temperature, and health context if they are to realize their full robustness potential. The benchmark is designed to be extensible, and the next steps are the extension to more cells and scenarios, the inclusion of remaining disturbance classes, and the transfer of the methodology toward broader embedded or hardware-in-the-loop validation.

Appendix A. Supplementary Result Tables

Table A.5: Baseline performance metrics for the selected benchmark trajectory.

Class	MAE	RMSE	P95 error	Max error	Bias
Direct measurement	0.0311	0.0397	0.0800	1.0000	0.0311
Hybrid direct measurement	0.0024	0.0061	0.0094	1.0000	0.0022
Hybrid ECM	0.0090	0.0127	0.0241	1.0000	-0.0029
Data-driven	0.0100	0.0143	0.0336	0.0686	-0.0062

Table A.6: Cross-scenario global performance metrics used in the result discussion.

Scenario	Class	MAE	ΔMAE	RMSE	ΔRMSE	P95 error
Current noise (high)	DM	0.0310	-0.0001	0.0394	-0.0002	0.0795
Current bias	DM	0.3143	0.2833	0.3683	0.3286	0.6670
Initial SOC error	DM	0.0322	0.0011	0.0413	0.0017	0.0833
Irregular sampling	DM	0.0309	-0.0001	0.0396	-0.0001	0.0799
Burst dropout	DM	0.0332	0.0022	0.0487	0.0090	0.0819
Missing samples	DM	0.0386	0.0076	0.0476	0.0079	0.0926
Voltage spikes	DM	0.0312	0.0001	0.0398	0.0001	0.0801
Temperature noise	DM	0.0311	0.0000	0.0397	0.0000	0.0800
Voltage noise	DM	0.0524	0.0213	0.0648	0.0252	0.1228
Current noise (high)	HDM	0.0044	0.0019	0.0074	0.0013	0.0120
Current bias	HDM	0.3068	0.3043	0.3638	0.3577	0.6668
Initial SOC error	HDM	0.0036	0.0011	0.0136	0.0075	0.0101
Irregular sampling	HDM	0.0027	0.0003	0.0061	0.0001	0.0095
Burst dropout	HDM	0.0051	0.0027	0.0325	0.0265	0.0098
Missing samples	HDM	0.0103	0.0079	0.0129	0.0069	0.0221
Voltage spikes	HDM	0.0062	0.0038	0.0140	0.0079	0.0328
Temperature noise	HDM	0.0067	0.0043	0.0110	0.0049	0.0210
Voltage noise	HDM	0.0751	0.0727	0.0908	0.0847	0.1578
Current noise (high)	HECM	0.0165	0.0075	0.0280	0.0154	0.0701
Current bias	HECM	0.0932	0.0842	0.1303	0.1176	0.2954
Initial SOC error	HECM	0.0091	0.0001	0.0143	0.0016	0.0241
Irregular sampling	HECM	0.0095	0.0004	0.0131	0.0004	0.0248
Burst dropout	HECM	0.0104	0.0014	0.0228	0.0101	0.0246
Missing samples	HECM	0.0119	0.0029	0.0163	0.0036	0.0329
Voltage spikes	HECM	0.0090	0.0000	0.0127	-0.0000	0.0241
Temperature noise	HECM	0.0090	0.0000	0.0127	0.0000	0.0241

Scenario	Class	MAE	ΔMAE	RMSE	ΔRMSE	P95 error
Voltage noise	HECM	0.0090	-0.0000	0.0127	-0.0000	0.0241
Current noise (high)	DD	0.0109	0.0009	0.0151	0.0008	0.0345
Current bias	DD	0.2859	0.2759	0.3426	0.3283	0.6367
Initial SOC error	DD	0.0110	0.0010	0.0175	0.0032	0.0355
Irregular sampling	DD	0.0105	0.0005	0.0144	0.0001	0.0330
Burst dropout	DD	0.0122	0.0022	0.0316	0.0173	0.0348
Missing samples	DD	0.0117	0.0016	0.0149	0.0006	0.0302
Voltage spikes	DD	0.0144	0.0044	0.0213	0.0070	0.0474
Temperature noise	DD	0.0181	0.0081	0.0223	0.0080	0.0451
Voltage noise	DD	0.0698	0.0597	0.0839	0.0696	0.1501

Table A.7: Local behavior and recovery metrics used in the result discussion.

Class	Scenario	Local metric	Value	Threshold
DM	Initial SOC error	Recovery time to strict band [h]	1.1658	0.0800
DM	Initial SOC error	Recovery time to fair band [h]	0.2500	0.1599
HDM	Initial SOC error	Recovery time to strict band [h]	–	0.0094
HDM	Initial SOC error	Recovery time to fair band [h]	–	0.0200
HECM	Initial SOC error	Recovery time to strict band [h]	2.3693	0.0241
HECM	Initial SOC error	Recovery time to fair band [h]	1.1564	0.0482
DD	Initial SOC error	Recovery time to strict band [h]	1.0464	0.0336
DD	Initial SOC error	Recovery time to fair band [h]	1.0014	0.0672
DM	Burst dropout	Recovery time after burst dropout [h]	27.3511	0.0216
HDM	Burst dropout	Recovery time after burst dropout [h]	27.4093	0.0008
HECM	Burst dropout	Recovery time after burst dropout [h]	28.5611	0.0095
DD	Burst dropout	Recovery time after burst dropout [h]	27.4351	0.0070
DM	Voltage spikes	Median spike recovery time [s]	0.0000	0.0800

Class	Scenario	Local metric	Value	Threshold
HDM	Voltage spikes	Median spike recovery time [s]	0.0000	0.0094
HECM	Voltage spikes	Median spike recovery time [s]	0.0000	0.0241
DD	Voltage spikes	Median spike recovery time [s]	0.0000	0.0336
DM	Current noise (high)	Late minus early excess rolling MAE	-0.0013	–
DM	Current noise (high)	Mean excess rolling MAE	-0.0001	–
HDM	Current noise (high)	Late minus early excess rolling MAE	-0.0011	–
HDM	Current noise (high)	Mean excess rolling MAE	0.0019	–
HECM	Current noise (high)	Late minus early excess rolling MAE	0.0038	–
HECM	Current noise (high)	Mean excess rolling MAE	0.0073	–
DD	Current noise (high)	Late minus early excess rolling MAE	-0.0002	–
DD	Current noise (high)	Mean excess rolling MAE	0.0009	–

Table A.8: Scenario overview linking disturbance configuration, primary evidence type, and main paper figure. This table clarifies which scenarios are interpreted predominantly through global metrics and which require additional local evidence.

Scenario	Configuration	Primary evidence	Main figure(s)
ADC quantization	Steps of 0.01 A, 0.005 V, and 0.5 °C	Global ΔMAE; detailed values in Appendix	Figs. 12, B.14
Current bias	Constant current offset in the compact matrix, plus dedicated 0.5%/1.5%/3.0% sweep of $\lvert I\rvert_{\max}$	Global MAE degradation across the sweep	Fig. 5

Scenario	Configuration	Primary evidence	Main figure(s)
Current noise	Compact matrix with $\sigma_I = 0.10\,\mathrm{A}$; local detail sweep up to $\sigma_I = 0.20\,\mathrm{A}$	Global ΔMAE plus local output-variability / rolling-MAE evidence	Fig. 6
Voltage noise	Additive Gaussian noise with $\sigma_U = 0.01\,\mathrm{V}$	Global ΔMAE	Fig. 12
Temperature noise	Additive Gaussian noise with $\sigma_T = 1.0\,^\circ\mathrm{C}$	Global ΔMAE	Fig. 12
Initial SOC error	Additive initial mismatch of -0.10 SOC; DD via online Q_c offset	Local recovery to strict and fair baseline bands; global MAE is secondary	Fig. 7
Missing samples	Every 50th sample frozen on the nominal 1 s grid	Global ΔMAE	Figs. 8, 12
Irregular sampling	Uniform sampling-time jitter of $\pm 0.1\,\mathrm{s}$ around the nominal 1 s grid	Global ΔMAE	Figs. 8, 12
Burst dropout	One central frozen gap of 3600 s	Global ΔMAE plus local transition and long-horizon recovery	Figs. 8, 9, 10
Voltage spikes	One $\pm 0.20\,\mathrm{V}$ single-sample spike every 1000 samples	Global ΔMAE plus aligned local excess-error response	Figs. 11, 12

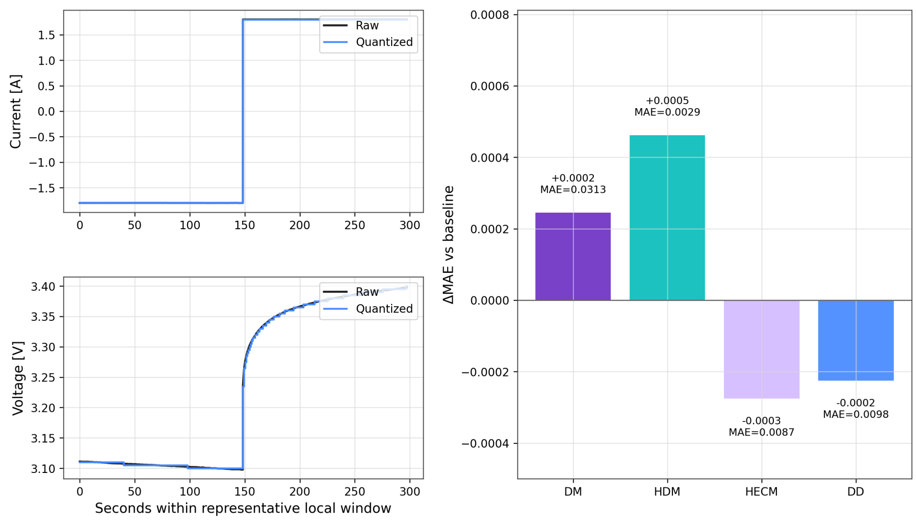

Figure B.14: ADC-quantization results. The two left panels illustrate representative current and voltage quantization in a local window, and the right panel summarizes the resulting global ΔMAE relative to the baseline.

Appendix B. ADC Quantization

ADC quantization is evaluated with step sizes of 0.01 A for current, 0.005 V for voltage, and 0.5 °C for temperature. Across all four estimator classes, the resulting deviations from the baseline remain small, so the detailed numerical comparison is collected here in the appendix while the scenario itself remains part of the overall benchmark.

Table B.9: ADC-quantization results for the four benchmarked estimator classes.

Class	Baseline MAE	ADC MAE	ΔMAE
DM	0.0311	0.0313	+0.0002
HDM	0.0024	0.0029	+0.0005
HECM	0.0090	0.0087	-0.0003
DD	0.0100	0.0098	-0.0002

Funding

Funding details are omitted for double-blind review.

Acknowledgements

Acknowledgements are omitted for double-blind review.

CRediT authorship contribution statement

The author contribution statement is provided in the non-blinded submission materials.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT, DeepL, and Grammarly in order to improve the linguistic quality and style of the manuscript. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Data availability

The underlying laboratory dataset is the MGFarm LFP 18650 cyclic-aging dataset described in [28]. The benchmark scripts and model runners are located in `DL\_Models/LFP\_SOC\_SOH\_Model/4\_simulation\_environment`, and the paper-facing figures and tables are collected under `DL\_Models/LFP\`

_SOC_SOH_Model/4_simulation_environment/results. The data and code underlying this study will be made available on reasonable request.

References

[1] P. Shrivastava, P. A. Naidu, S. Sharma, B. K. Panigrahi, A. Garg, Review on technological advancement of lithium-ion battery states estimation methods for electric vehicle applications, Journal of Energy Storage 64 (2023) 107159. doi:10.1016/j.est.2023.107159.

[2] F. Guo, L. D. Couto, G. Mulder, K. Trad, G. Hu, O. Capron, K. Haghverdi, A systematic review of electrochemical model-based lithium-ion battery state estimation in battery management systems, Journal of Energy Storage 101 (2024) 113850. doi:10.1016/j.est.2024.113850.

[3] J. Yao, J. Kowal, Towards a smarter battery management system: A critical review on deep learning-based state of charge estimation of lithium-ion batteries, Energy and AI 21 (2025) 100585. doi:10.1016/j.egyai.2025.100585.

[4] X. Li, J. Jiang, C. Zhang, L. Y. Wang, L. Zheng, Robustness of SOC Estimation Algorithms for EV Lithium-Ion Batteries against Modeling Errors and Measurement Noise, Mathematical Problems in Engineering 2015 (2015) 1–14. doi:10.1155/2015/719490.

[5] F. Naseri, C. Barbu, A. Jensen, P. Setoodeh, J. R. Baena, N. Hevele, E. Schaltz, Toward Reliable Wireless BMS: Robust Battery State Estimation Under Noise and Uncertainty, IEEE Transactions on Vehicular Technology (2025) 1–12doi:10.1109/TVT.2025.3638613.

[6] F. Berger, D. Joest, E. Barbers, K. Quade, Z. Wu, D. U. Sauer, P. Dechent, Benchmarking battery management system algorithms - Requirements, scenarios and validation for automotive applications, eTransportation 22 (2024) 100355. doi:10.1016/j.etran.2024.100355.

[7] Z. A. Khan, P. Shrivastava, S. M. Amrr, S. Mekhilef, A. A. Algethami, M. Seyedmahmoudian, A. Stojcevski, A Comparative Study on Different Online State of Charge Estimation Algorithms for Lithium-Ion Batteries, Sustainability 14 (12) (2022) 7412. doi:10.3390/su14127412.

[8] Y. Gao, G. L. Plett, G. Fan, X. Zhang, Enhanced state-of-charge estimation of LiFePO4 batteries using an augmented physics-based model, Journal of Power Sources 544 (2022) 231889. doi:10.1016/j.jpowsour.2022.231889.

[9] Y. Gao, K. Liu, C. Zhu, X. Zhang, D. Zhang, Co-Estimation of State-of-Charge and State-of- Health for Lithium-Ion Batteries Using an Enhanced Electrochemical Model, IEEE Transactions on Industrial Electronics 69 (3) (2022) 2684–2696. doi:10.1109/TIE.2021.3066946.

[10] J. Shen, W. Ma, J. Xiong, X. Shu, Y. Zhang, Z. Chen, Y. Liu, Alternative combined co-estimation of state of charge and capacity for lithium-ion batteries in wide temperature scope, Energy 244 (2022) 123236. doi:10.1016/j.energy.2022.123236.

[11] C. Wang, N. Cui, Z. Cui, H. Yuan, C. Zhang, Fusion estimation of lithium-ion battery state of charge and state of health considering the effect of temperature, Journal of Energy Storage 53 (2022) 105075. doi:10.1016/j.est.2022.105075.

[12] P. Mondal, D. Bhavsar, K. Mittal, M. Mittal, Estimating State-of-Charge in Lithium-Ion Batteries Through Deep Learning Techniques: A Comparative Evaluation, IEEE Access 12 (2024) 78773–78786. doi:10.1109/ACCESS.2024.3408220.

[13] D. P. Pau, A. Aniballi, Tiny Machine Learning Battery State-of-Charge Estimation Hardware Accelerated, Applied Sciences 14 (14) (2024) 6240. doi:10.3390/app14146240.

[14] S. Giazitzis, M. Sakwa, S. Leva, E. Ogliari, S. Badha, F. Rosetti, A Case Study of a Tiny Machine Learning Application for Battery State-of-Charge Estimation, Electronics 13 (10) (2024) 1964. doi:10.3390/electronics13101964.

[15] Y. Mazzi, H. Ben Sassi, A. Gaga, F. Errahimi, State of charge estimation of an electric vehicle's battery using tiny neural network embedded on small microcontroller units, International Journal of Energy Research 46 (6) (2022) 8102–8119. doi:10.1002/er.7713.

[16] M. A. Kamali, A. C. Caliwag, W. Lim, Novel SOH Estimation of Lithium-Ion Batteries for Real-Time Embedded Applications, IEEE Embedded Systems Letters 13 (4) (2021) 206–209. doi:10.1109/LES.2021.3078443.

[17] B. Gou, Y. Xu, X. Feng, An Ensemble Learning-Based Data-Driven Method for Online State-of-Health Estimation of Lithium-Ion Batteries, IEEE Transactions on Transportation Electrification 7 (2) (2021) 422–436. doi:10.1109/TTE.2020.3029295.

[18] W. Li, N. Sengupta, P. Dechent, D. Howey, A. Annaswamy, D. U. Sauer, Online capacity estimation of lithium-ion batteries with deep long short-term memory networks, Journal of Power Sources 482 (2021) 228863. doi:10.1016/j.jpowsour.2020.228863.

[19] F. Rzepka, P. Hematty, M. Schmitz, J. Kowal, Neural Network Architecture for Determining the Aging of Stationary Storage Systems in Smart Grids, Energies 16 (17) (2023) 6103. doi:10.3390/en16176103.

[20] BQ79616 16-Series Battery Monitor, Balancer, and Integrated Hardware Protector (2023).

[21] Design Considerations of Current Sensing With BQ769x2 Family for Battery Management System (2022).

[22] TMCS1108 3% Functional Isolation Hall-Effect Current Sensor With ±100-V Working Voltage (2024).

[23] TLE4972 high precision coreless current sensor for automotive applications (2023).

[24] O. Aiello, Electromagnetic Susceptibility of Battery Management Systems' ICs for Electric Vehicles: Experimental Study, Electronics 9 (3) (2020) 510. doi:10.3390/electronics9030510.

[25] M. Liu, D. Geißler, S. Bian, B. Zhou, P. Lukowicz, Assessing the Impact of Sampling Irregularity in Time Series Data: Human Activity Recognition As A Case Study (Jan. 2025). arXiv:2501.15330, doi:10.48550/arXiv.2501.15330.

[26] 10s battery pack monitoring, balancing, and comprehensive protection, 50-A discharge reference design (2024).

[27] A. Aqachtoul, K. Karam, A. Elamrani, Y. Alidrissi, M. Najib, A. Rochd, M. Bakhouya, MGFARM: An IoT and Edge-Driven Framework for Smart and Sustainable Agricultural Practices, in: Y. Mejdoub, A. Elamri, M. Kardouchi (Eds.), Connected Objects, Artificial Intelligence, Telecommunications and Electronics Engineering, Vol. 1584, Springer Nature Switzerland, Cham, 2026, pp. 419–425. doi:10.1007/978-3-032-01536-5_64.

[28] F. Rzepka, Design of Experiments for Cyclic Aging of LFP 18650 Cells: Impact of Charge/Discharge C-Rates, Depth of Discharge, and Temperature on Battery Degradation (Nov. 2025). doi:10.14279/DEPOSITONCE-24800.

[29] K. Siebertz, D. Van Bebber, T. Hochkirchen, Statistische Versuchsplanung, Springer Berlin Heidelberg, Berlin, Heidelberg, 2017. doi:10.1007/978-3-662-55743-3.